

MAJOR HAZARD REPORT - STRING PACK-OFF & NPT

SYNTHETIC DEMONSTRATION DOCUMENT - NOT REAL OPERATIONAL DATA (SIH PROTOTYPE)

WELL IDENTIFIER: NWIS-W008

FORMATION: Demo-Kopili Shale

EVENT TYPE: STUCK_PIPE

OPERATIONAL DATE: 2024-05-28

MEASURED DEPTH: 3560.0 m

SEVERITY LEVEL: CRITICAL

OPERATIONAL NARRATIVE

While reaming tight hole section at 3560m in deep Demo-Kopili Shale, severe mechanical pack-off occurred. Standpipe pressure spiked to 3800 psi (pressure relief valve triggered). Rotation stalled completely.

CAUSE INVESTIGATION

Deep overpressured Kopili marine shale spalling due to insufficient mud weight (1.28 SG). Severe borehole ovalization and stress-induced spalling created thick debris bed around stabilizers.

MITIGATION & RECOVERY

- Disengaged top drive; applied maximum allowable upward jar impacts (75 tons overpull).
- Spotted concentrated acid wash followed by synthetic oil-based lubricant pill.
- Drillstring freed after 14 hours of jarring; total 32 hours Non-Productive Time (NPT) logged.
- Raised drilling mud weight to 1.38 SG to provide adequate mechanical borehole support.